Prototyping

Busem Deniz

Version 1.0 · Academic Year 2025/26 · Prototyping Course

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ABOUT ME

Busem Deniz

Multimedia & Creative Technologies student with a strong passion for video games and interactive media. I enjoy combining creativity and technical skills to create impactful experiences, with the goal of entering the game industry specializing in narrative design and/or 3D digital production.

Outside of university, I explore creative tools independently, work on personal projects, and study storytelling across games, film, books, and obline media. I am particularly drawn to projects that use environmental details, sound, and aesthetic design to communicate emotionally and psychologically driven narratives.

SKILLS

Unity • Blender • Illustrator • Premiere Pro • After Effects • Audacity • FreeCAD • Visual Studio Code • Arduino • Laser Cutting • 3D Printing

INTERESTS

FPS Games • Horror & Physchological Media • Philosophy • Worldbuilding • 3D Modeling • Video Editing • Motion Graphics • Concept Development

PROJECT BRIEF

Aura Light Billboard

01

OUTCOME(S)

- 4x4x4 cm paper cube prototype
- Digital CubeLight poster design
- Blender advertisement scene
- Printed paper prototype

TOOLS

- A4 paper
- Stationery (scissors, tape, ...)
- Canva
- Blender

DURATION

~ 1 week

SOLO OR TEAM

Solo project

CONTEXT

This project introduced the fundamentals of prototyping through a simple but precise construction exercise. The task was to create a 4x4x4 cm cube from a single sheet of paper and later recreate the object digitally in Blender. By combining physical crafting with digital reconstruction, the assignment explored the relationship between real-world geometry and 3D modelling workflows.

The project served as the foundation for the following stages, where the cube would evolve into a complete product concept and advertisement campaign.

LEARNING OBJECTIVES

- Develop an understanding of prototyping workflows through both physical construction and digital modelling techniques.
- Learn the fundamentals of Blender, including modelling, lighting, rendering, environmental composition, and paper model exporting.
- Combine physical crafting, digital design, rendering, and assembly techniques into one complete creative production pipeline.

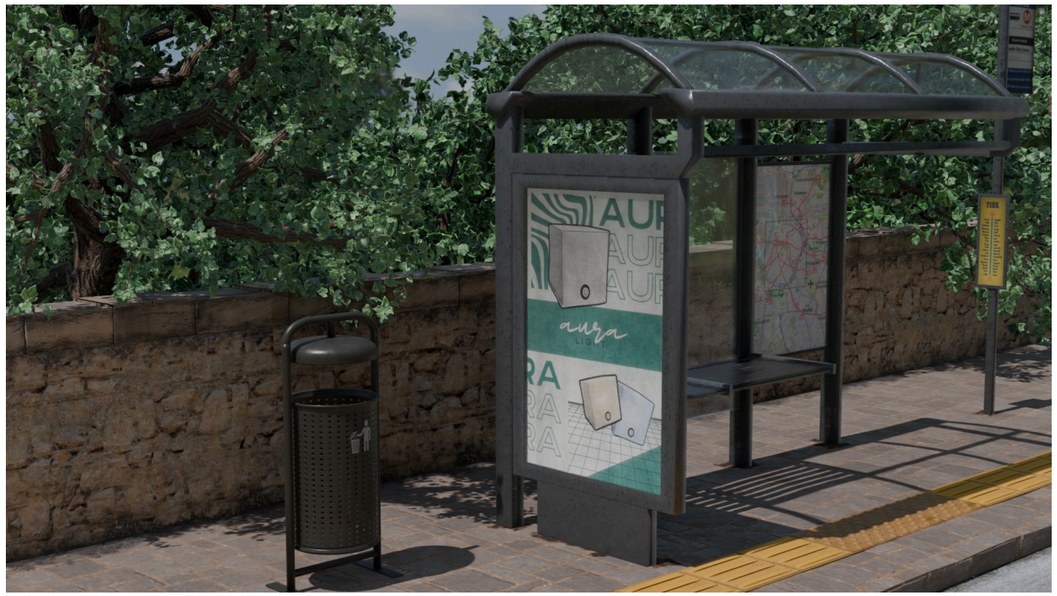
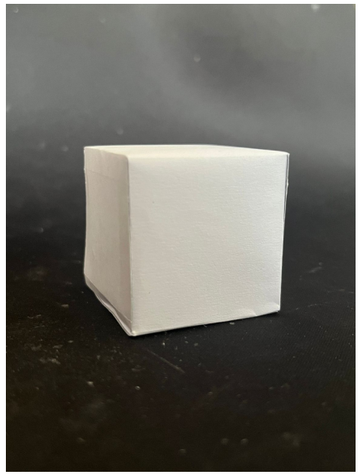
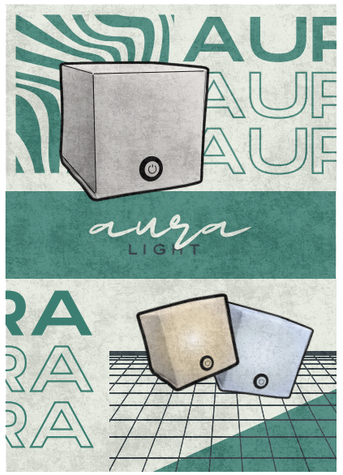
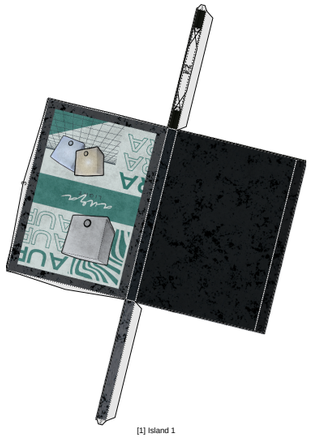


Figure 1: Bus stop scene featuring "Aura Light", made in Blender.







Figures 2, 3 & 4: From left to right, A4 paper cube, 4x4x4; "Aura Light" promotional poster, made in Canva; PDF export of billboard cutout, made in Blender.

PROTOTYPING PROCESS

The project began with a timed prototyping exercise where I constructed a 4x4x4 cm cube from a single sheet of A4 paper using basic stationery materials. I first sketched six connected 4x4 squares with additional side flaps before cutting, folding, and temporarily assembling the structure with tape. Once the proportions and overall form felt refined, the cube was permanently glued together.

After completing the physical prototype, I transformed the cube into a conceptual product called “Aura Light,” a minimalist paper lamp focused on atmosphere and mood. The concept features a single integrated button that cycles through cold, neutral, and warm lighting temperatures, allowing the ambience to change depending on the user’s mood.

Later, I developed a A3 size promotional poster in Canva, designed around calming and spiritual aesthetics using a sage green and beige colour palette combined with organic swirl patterns. Sharp black outlines and overlaid paper textures were added to emphasize the handcrafted paper material and reinforce the physical origins of the prototype.

To expand the project beyond the physical prototype and poster design, I recreated the Aura Light campaign inside a realistic 3D bus stop environment in Blender. The goal was to present the advertisement within a believable public setting while exploring environmental modelling, texturing, lighting, and rendering workflows.

The scene was designed with a strong focus on realism and environmental storytelling. Multiple small surface details were added to avoid a clean or artificial appearance. Dirt and weathering textures were placed around the connections between the pavement and wall surfaces, while subtle rust buildup was added around the metal supports and structural joints of the bus stop to suggest age and long-term outdoor exposure.



Figures 5a, 5b, 5c & 5d: Bus stop scene close-up shots, focusing on different environmental details.

The pavement was carefully modelled with separated material transitions between the pedestrian walkway and roadway edge. The tactile guidance strips for visually impaired pedestrians were modelled with raised geometry rather than relying only on flat textures, allowing light and shadows to interact naturally with the surface details during rendering.

To create a more atmospheric setting, dense vegetation and overhanging trees were added behind the stone wall, helping frame the advertisement while introducing contrast between the organic environment and the industrial bus stop structure. Additional environmental elements such as the Antwerp transit map, bus schedule signage, glass panels, trash bin, and realistic shadow casting were included to strengthen the realism of the scene.

The final renders aimed to merge the physical prototype, branding work, and environmental visualization into one cohesive presentation while demonstrating attention to material detail, composition, and realistic world-building.

After completing the digital model in Blender, I used the Export Paper Model add-on to convert the 3D object into a printable paper template. The add-on unfolded the cube geometry into flat connected surfaces that could be printed, cut out, folded, and reassembled physically.

This process introduced the relationship between digital geometry and real-world construction constraints. Careful attention had to be paid to edge placement, folding tabs, and assembly order to ensure the model could be reconstructed accurately once printed. After exporting the template, the design was printed onto paper, manually cut out, folded along the generated seams, and assembled into a physical version of the model.

The workflow demonstrated how digital 3D models can be translated back into tangible prototypes while preserving form and proportions. It also reinforced the importance of precision in modelling, since even small inaccuracies in the digital structure could affect the physical assembly process.

PROJECT BRIEF

FreeCAD Part Creation 02

OUTCOME(S)

- Two fully modelled mechanical parts recreated in FreeCAD
- Exported PDF presentation sheets of the completed models

TOOLS

- FreeCAD
- Technical drawing reference sheets

DURATION

2 days

SOLO OR TEAM

Solo project

CONTEXT

The purpose of this assignment was to develop an understanding of technical drawing interpretation and precision modelling workflows commonly used in engineering and product development. Students were expected to analyze professional mechanical drawing sheets and recreate the objects digitally while maintaining correct proportions, dimensions, and structural details.

The assignment focused on precision and technical accuracy by requiring students to recreate mechanical parts using exact dimensions, sectional views, and geometric details. Instead of conceptual or artistic modelling, the project emphasized clean construction methods and problem-solving through CAD workflows.

LEARNING OBJECTIVES

- Develop the ability to interpret orthographic projections, sectional views, and technical dimensions.
- Learn the fundamentals of CAD modelling workflows inside FreeCAD, including sketching, constraints, extrusion, and boolean operations.
- Improve precision and spatial reasoning skills through the reconstruction of mechanical parts from 2D references.

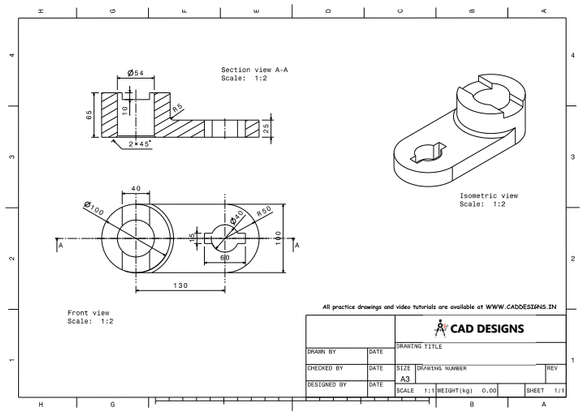


Figure 6: Model 11 reference sheet.

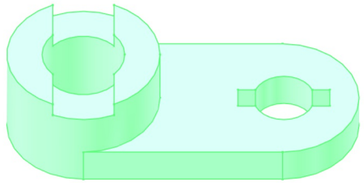


Figure 7: PDF export of model 11 on FreeCAD.

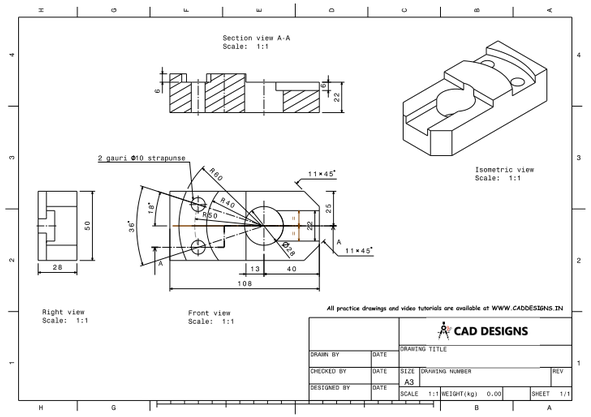


Figure 8: Model 19 reference sheet.

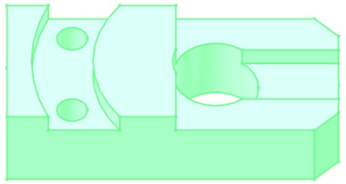


Figure 9: PDF export of model 19 on FreeCAD.

PROTOTYPING PROCESS

The project began with carefully studying the selected technical drawings and identifying the main geometric forms that structured each object. Before modelling, I analyzed the orthographic projections, sectional views, dimensions, radiuses, and cut sections to better understand how the separate technical views translated into complete 3D forms. Since the project focused on engineering accuracy, understanding the drawings correctly before starting was an important part of the workflow.

Inside FreeCAD, both parts were constructed through constrained sketches, extrusion operations, circular cuts, chamfers, and boolean workflows. I started with the primary base shapes before adding smaller structural details such as cylindrical elevations, recessed sections, rounded edges, and drilled openings. Maintaining proper alignment and proportions throughout the process required constant comparison with the original technical sheets.

One of the main challenges during the modelling process was working with constraints and sketch relationships inside FreeCAD. Incorrectly constrained sketches occasionally caused geometry warnings or prevented certain operations from functioning correctly. In some cases, small inaccuracies in measurements or misplaced constraints affected later extrusion and cutting steps, forcing me to revisit earlier parts of the model and rebuild sections more cleanly.

Another difficulty was interpreting certain sectional and orthographic views spatially. Some shapes appeared straightforward in one perspective but became more complex when reconstructed three-dimensionally, requiring continuous comparison between different viewpoints during the modelling process.

After completing the models, the final parts were exported as PDF documents. Overall, the project reinforced the importance of precision, planning, and structured CAD workflows when translating 2D engineering drawings into accurate digital models.

PROJECT BRIEF

Ping Pong Launcher

03

OUTCOME(S)

- Documentation of prototyping and fabrication process in PDF
- Functional ping pong launcher
    - Arduino-controlled electronic system
    - Laser-cut and 3D-printed structural components

TOOLS

- Cardboard
- Laser cutter (Adobe Illustrator)
- 3D Printer (FreeCAD)
- Soldering tools & electronic components
- Arduino UNO & Arduino IDE

DURATION

~ 4 weeks

SOLO OR TEAM

Solo project

CONTEXT

This project focused on the development of a functional ping pong launcher through a combination of mechanical prototyping, electronics, and digital fabrication. The project challenged students to design a machine capable of launching a ping pong ball approximately three meters into a bucket while maintaining stability, repeatability, and safe operation. The prototype was required to combine laser-cut and 3D-printed components alongside an Arduino-controlled system, encouraging an iterative workflow built around testing, adjustment, and technical problem-solving.

Unlike previous projects centered mainly around modelling or visual presentation, this assignment emphasized functionality, system integration, and the relationship between physical construction and electronic interaction.

LEARNING OBJECTIVES

- Develop an understanding of iterative prototyping through testing, problem-solving, and refinement.
- Learn how to integrate Arduino electronics with mechanical systems and motor control.
- Apply digital fabrication techniques such as laser cutting, 3D printing, and CAD-based design.
- Improve structural and spatial design skills through enclosure planning and component integration.



Figure 10: "BEARCLAW" front view.

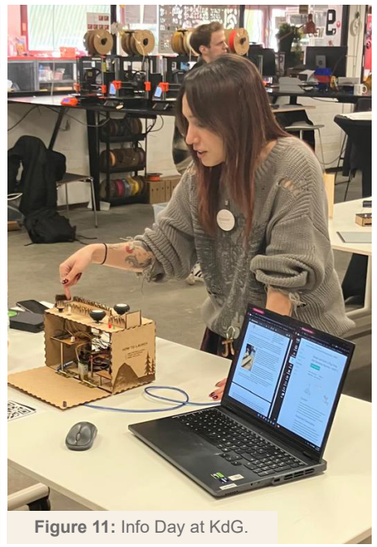


Figure 11: Info Day at KdG.

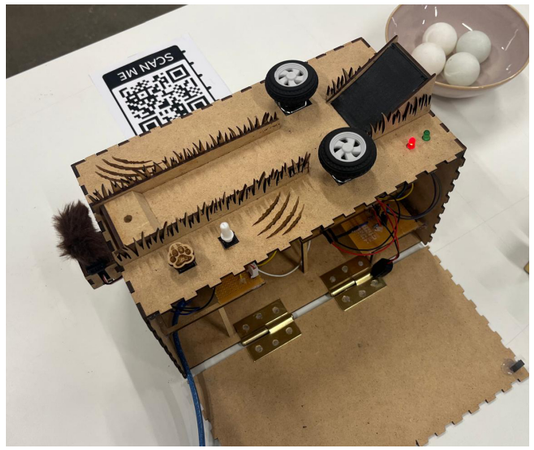


Figure 12a: Top view, showcasing the launcher surface.

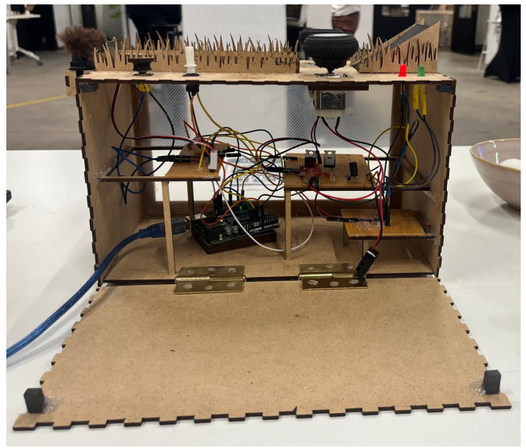


Figure 12b: Back view, showcasing the cable management.

PROTOTYPING PROCESS

The development process of the "BEARCLAW" ping pong launcher was highly iterative and focused on testing mechanical and electronic behavior before committing to permanent fabrication. Rather than building the final version immediately, I approached the project through multiple prototype stages that exposed structural weaknesses, electrical limitations, and interaction problems early in development. The project evolved through experimentation, redesigns, and several major pivots before reaching a stable final system.

The first prototype explored a fully mechanical elastic-powered launcher using PVC tubes, an elastic band, and a manually pulled plunger system. This early version was created to test whether sufficient force and repeatability could be achieved without electronics. While the mechanism functioned, testing quickly revealed major limitations such as short acceleration distance, friction between the PVC tubes, unstable alignment, and inconsistent launch force caused by manual input. The launcher was only capable of reaching around 40% of the required range, making it unsuitable for the assignment goals. Instead of forcing further improvements onto a structurally flawed system, I decided to abandon the concept entirely and pivot toward a motor-driven approach.

The second stage focused on developing a dual-wheel flywheel launcher digitally in Blender before physical fabrication. During this phase, I explored how ping pong balls could be accelerated consistently using rotating wheels rather than elastic force. The first concept included a servo-controlled lifting platform, but while sketching and modelling the mechanism, I realized the design introduced unnecessary complexity and increased the risk of jamming and alignment errors. This led to simplifying the interaction into a pushing mechanism instead, which became the basis of the final system.

A large portion of the prototyping process involved troubleshooting electronics and subsystem integration.

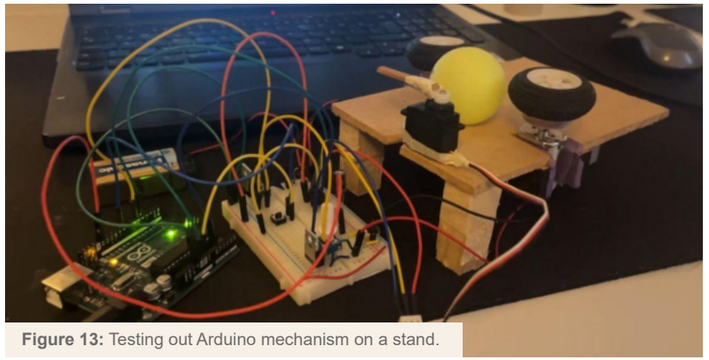


Figure 13: Testing out Arduino mechanism on a stand.

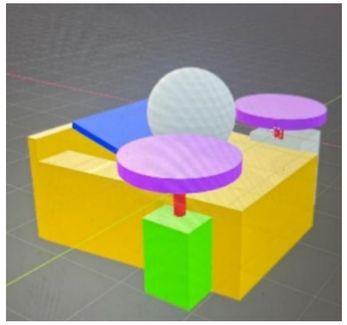


Figure 14: New concept model (Blender).

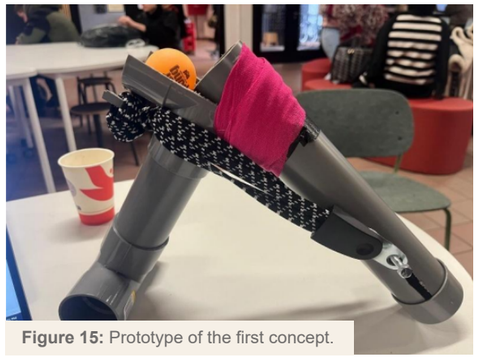


Figure 15: Prototype of the first concept.

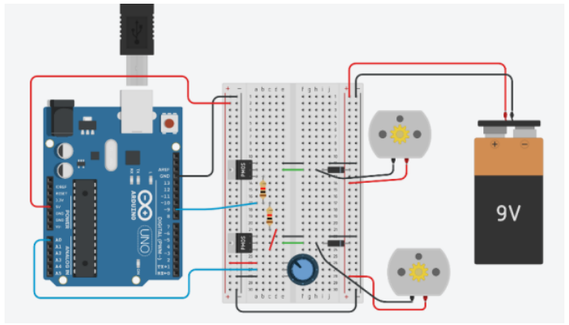


Figure 16: DC motors and potentiometer wiring (Tinkercad).

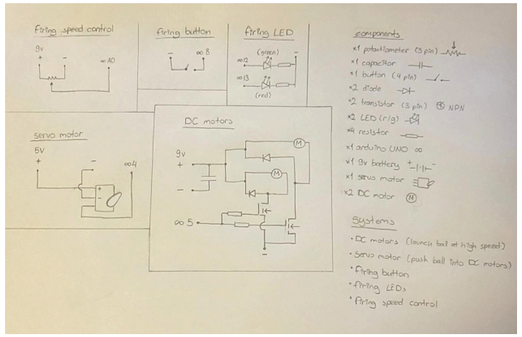


Figure 17: Schematics of seperate circuits.

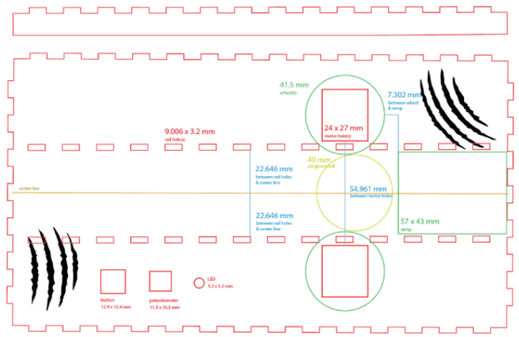


Figure 18: Top surface component placements (Illustrator).

I began by testing a single DC motor with Arduino PWM control and later expanded the setup into a dual-motor system. One of the first major problems appeared when both motors were connected in parallel, causing inconsistent behavior where only one motor functioned reliably. After debugging the issue, I discovered that each motor required its own transistor and flyback diode for proper isolation and stable current flow. Later, I also encountered problems with motor direction, vibration, unstable wheel alignment, voltage drops, servo interference, and noisy power delivery. Many of these issues only became visible through physical testing and repeated iteration.

To solve these challenges, I prototyped every subsystem independently before integrating them together. Breadboard testing allowed rapid experimentation with the button logic, LEDs, potentiometer control, servo behavior, and DC motors separately before combining them into a complete architecture. Mechanical prototyping followed a similar philosophy. Temporary wooden stands, cardboard bases, taped servo mounts, MDF test pieces, and improvised rails were repeatedly adjusted to test spacing, wheel compression, ramp angles, and ball guidance before permanent fabrication began. Some prototypes initially appeared successful but later failed when transferred to rigid materials, forcing me to reevaluate assumptions and redesign parts of the structure.

As the project progressed, prototyping shifted from purely functional testing toward refinement and integration. Laser-cut MDF structures, 3D-printed ramps, custom FreeCAD motor couplers, modular perfboards, hinged access doors, magnetic closures, and themed visual components were all developed through iterative testing and adjustment. Each stage exposed new practical problems such as vibration, structural instability, cable routing conflicts, or insufficient momentum transfer, requiring further redesigns before reaching a stable final result.

The final launcher became the product of continuous evaluation and incremental problem-solving rather than a single predefined idea executed from start to finish.

Watch the display video here.

PROJECT BRIEF

Shadow Theatre

04

OUTCOME(S)

- Documentation of prototyping and fabrication process in PDF
- Arduino-powered interactive puppet theatre
    - Integrated lighting, sound, and LCD interaction systems
    - Laser-cut MDF stage and controller enclosures

DURATION

~ 6 weeks

TOOLS

- Cardboard and A2 paper
- Baking paper
- Laser & vinyl cutter (Adobe Illustrator)
- Soldering tools & electronic components
- Arduino Mega & Arduino IDE

SOLO OR TEAM

Solo project

CONTEXT

This project focused on designing and fabricating an interactive prototype through iterative development, digital fabrication, and electronics integration. The assignment emphasized experimentation, technical execution, and documentation while encouraging the use of fabrication technologies such as laser cutting, enclosure design, and Arduino systems.

For this project, I designed and fabricated an interactive miniature puppet theatre that combined Arduino electronics, laser cutting, lighting systems, sound, and physical storytelling into one cohesive experience. The final concept evolved into a story-driven theatre where players interact with a custom controller to select characters, trigger puppet movement, activate lighting effects, and progress through short narrative sequences.

LEARNING OBJECTIVES

- Apply an iterative prototyping workflow through testing, redesigning, and documenting different development stages.
- Improve problem-solving skills by troubleshooting mechanical, structural, and electronic issues during development.
- Create a functional and reproducible prototype while maintaining clear documentation of the process and technical setup.
- Explore creative interaction design by combining movement, electronics, and themed visual presentation into a single product.



Figure 19a: Theatre stage front & top view, components detached.



Figure 19b: Doll props.

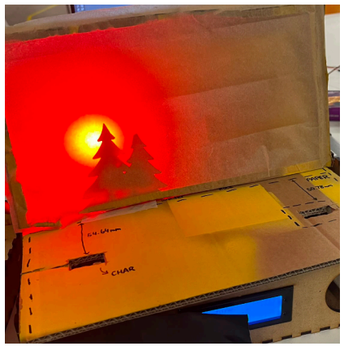


Figure 20: Light show display.

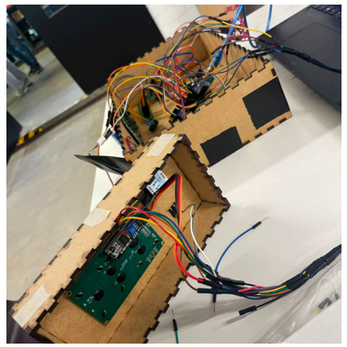


Figure 21: Arduino setup inside closures.

PROTOTYPING PROCESS

The prototyping process focused on validating the interaction systems, enclosure construction, lighting behavior, and mechanical reliability before permanently assembling the final theatre. Rather than directly building the final version, the project evolved through multiple prototype stages involving cardboard mockups, Arduino testing setups, paper measurements, MDF fabrication tests, and iterative redesigns.

The first stage focused entirely on Arduino functionality without integrating final aesthetics or story content. Servos, LEDs, MP3 playback, the LCD screen, rotary encoder, and button controls were all tested separately before being combined into one functional system. Temporary cardboard puppets were attached onto servo motors to test whether the rotating character-selection mechanism would function reliably and remain visually understandable from the audience perspective.

After confirming that the electronics worked together, prototyping shifted toward enclosure development. A cardboard controller prototype was built first to determine spacing for the LCD screen, rotary encoder, buttons, wiring, and Arduino components before producing the laser-cut MDF enclosure. Since the dimensions had already been validated physically, the final controller enclosure assembled without major issues.

The stage enclosure required significantly more experimentation due to the layered construction and lighting systems. A cardboard prototype was created to test servo placement, prop visibility, LED positioning, and spacing between different stage layers. Measurements and placement plans were drawn directly onto a yellow cardboard surface layer to determine where props, lighting, and moving elements could fit without interfering with each other.

One of the largest experimental phases involved testing the shadow-theatre lighting system using baking paper, LEDs, and layered props. Initial tests using the standard Arduino LEDs failed because the light was too weak to become visible through the baking

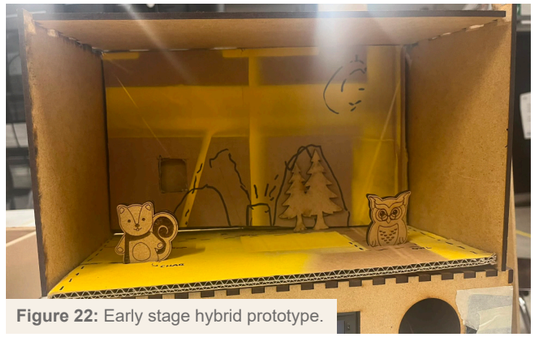


Figure 22: Early stage hybrid prototype.

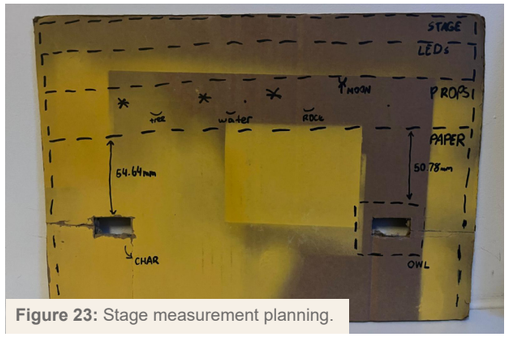


Figure 23: Stage measurement planning.

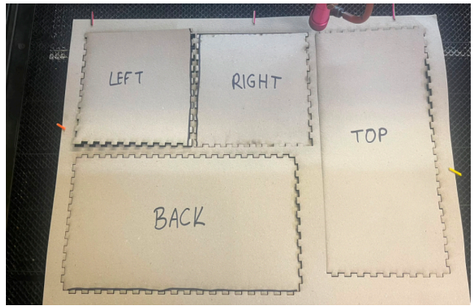


Figure 24: Laser-cut theatre enclosure prototype, A2 paper.

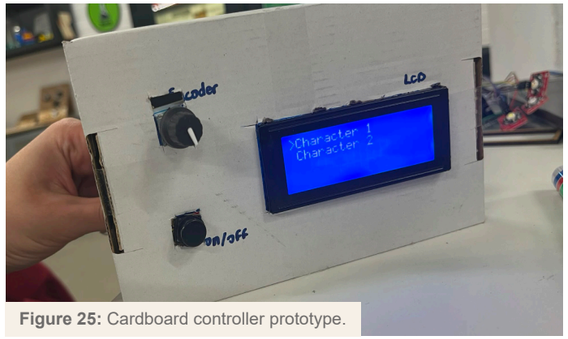


Figure 25: Cardboard controller prototype.





Figures 26a & 26b: Vinyl-cut controller graphics and theatre branding, illustrated in Illustrator.

paper. To improve visibility, the baking paper was attached directly onto the props themselves, but the effect still remained too subtle.

After replacing the LEDs with significantly stronger versions, a second issue appeared: individual LEDs became powerful enough to illuminate multiple nearby props simultaneously. To solve this, small hollow cardboard structures were built around each LED to isolate the light direction and ensure that each light source illuminated only one prop at a time. This solution successfully restored clarity and improved the readability of the stage scenes.

The MDF stage enclosure also went through several redesigns. The first lower enclosure only included openings for speakers, LEDs, and cable routing. Servo openings were tested separately on scrap MDF before being integrated into the final stage surface. Multiple paper test cuts were produced before cutting MDF in order to verify sizing and alignment.

The upper stage structure caused repeated measurement problems during fabrication. Despite several decimal-based adjustments inside Illustrator, the upper enclosure repeatedly became slightly too large or too small when connected to the lower stage. To solve this, the connection structure between both sections was redesigned by integrating additional interlocking gaps and modifying how the upper stage aligned with the lower enclosure. These adjustments were first tested on paper before producing the corrected MDF version.

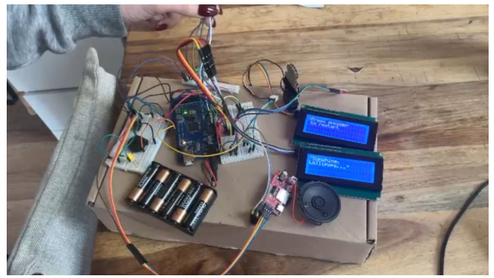
As the technical systems became stable, the project shifted toward aesthetic refinement. Laser engraving was added onto previously fabricated MDF pieces to avoid wasting additional material. Logos, decorative graphics, and environmental details were engraved directly onto the stage and controller enclosure. Vinyl cutting was also explored to add highlighted graphic elements and improve presentation quality.

The environmental framing of the stage evolved throughout the prototyping process as well. Traditional theatre curtains were initially considered, but testing revealed that they blocked visibility of both the puppets and illuminated props.

Instead, laser-cut tree structures were used to frame the stage opening while preserving visibility and reinforcing the atmosphere of the environment.

The prop frame system also changed during development. Early versions attempted to attach props directly onto the stage walls, but this proved difficult to adjust and maintain consistently. A more modular construction system was developed in which props and frame pieces were laser cut separately. The frame was divided into smaller pieces rather than one large structure, allowing easier sanding, resizing, and assembly whenever measurement inconsistencies appeared during fabrication.

At the current stage of development, the project contains a fully functional Arduino setup, synchronized servo and lighting systems, working audio playback, modular stage structures, laser-cut props, puppet mechanisms, and completed enclosure designs. The remaining work focuses primarily on integrating the final narrative and dialogue systems into the Arduino code before permanently assembling the theatre.



Watch the Arduino display here.



Watch the enclosure display here.

2025-2026